1 .Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



2 List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

| No. | Device | Network Type |
|---|---|---|
| 1 | Smartphone | WAN |
| 2 | Smart TV | LAN |
| 3 | Wi-Fi Router | LAN |
| 4 | Laptop | LAN |
| 5 | Smartwatch | PAN |

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

| Internet | Intranet |
|---|---|
| 1. The Internet is a worldwide network. | 1. An intranet is a private network. |
| 2. It connects computers and mobile phones around the world. | 2. It connects computers within an organization. |
| 3. Anyone can use the Internet with an Internet connection. | 3. Only authorized users can access an intranet. |
| 4. We use it to watch YouTube, use WhatsApp, and search for information. | 4. It is used to share private information, files, and college results. |
| 5. Example: Using public Wi-Fi to watch YouTube. | 5. Example: Using a private college portal to check attendance and assignments. |

4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

***Hint:*** *Mention the devices and networks involved, and what happens when you hit 'search'*.

- **Ex : WhatsApp**

✓ I open WhatsApp on my smartphone and type something in the search box.

✓ My smartphone connects to the Wi-Fi router or mobile network.

✓ The data travels through the ISP (Internet Service Provider) to the WhatsApp server.

✓ The WhatsApp server processes the request and sends the matching results back to my phone.

✓ My phone receives the data and displays the search results on the screen.